

# A 2.4 GHz Doherty Power Amplifier Based on Voltage Combining in 22 nm CMOS

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**Abstract**—In this paper, a 2.4 GHz Doherty power amplifier (PA) based on voltage combining in 22 nm CMOS is presented, which achieves high efficiency and linearity. The main and the auxiliary amplifier (aux. amplifier) both use differential cascode structure for high output power without reliability issues. The asymmetric structure is used to get high linearity. At 2.4 GHz, the simulated saturated output power, gain and 1 dB output power compression point are 30.7 dBm, 20.1 dB, and 28.1 dBm, respectively. The peak power added efficiency (PAAE) is 42.5 % by using 3.3 V supply voltage, and the PAAE at 6 dB back-off power is 25.2%, achieving the back-off power efficiency enhancement. The third order intermodulation distortion (IMD3) at 9 dB and 12 dB back-off powers are – 44.2 dBc and – 55.3 dBc. The Doherty PA designed in this paper is suitable for applications requiring high efficiency and linearity.

**Keywords**—CMOS power amplifier (PA), Doherty PA, voltage combining network, transformer, efficiency, linearity

## I. INTRODUCTION

The power amplifier (PA) is the most critical component of the wireless communication system, located at the end of the transmitter, and the study of the RF PAs has been a hot issue.

With the development of wireless communication technology, it is vital that PAs have high efficiency and high linearity in order to maximize the data rate in the fast moving environment. High efficiency and high linearity have always been the main contradiction in designing PAs, so a trade-off needs to be made in the design, which makes designing PAs more challenging.

To improve linearity, PA is mainly operated at power-back-off (PBO) levels, but the efficiency of PA decreases rapidly due to the fixed supply voltage and the load impedance optimized at the maximum power level [1]. This is one of the reasons why the PA must have a high PBO efficiency. In addition, the PA must have a high PBO efficiency due to the use of high peak-average-power-ratio (PAPR) orthogonal frequency-division multiplexing (OFDM) modulation [2]. The PA with high efficiency can extend the service life of the equipment. Therefore, it is very important to achieve a high PBO efficiency for the PAs. Improving the efficiency of the PAs at PBO levels has become an important focus of research in recent years.

In recent years, adaptive bias circuits, capacitor compensation [3], and predistortion [4] have been proposed to significantly improve the linearity of PAs. Envelope Elimination and Restoration (EER), Envelope Tracking (ET) and Doherty PAs have been proposed to improve the efficiency of PAs. In the EER technique, the original input signal contains both amplitude and phase information. The signal should be split into the envelope path and the RF path

containing the phase-modulated signal [5]. In the ET technique, the efficiency is improved by employing a dynamic supply voltage [6]. Although the former two systems can effectively improve the efficiency of the PA, they have some problems to be solved, such as the difficulty of adjusting the delay for the RF and envelope paths and the complexity of linkage between the PA and bias modulator [7]. The Doherty PA can accomplish the high efficiency at PBO levels through load modulation and two PAs with different bias. The Doherty PA is widely used in communication base stations because of its simple structure and easy implementation.

In this paper, the detailed implementation of the designed Doherty PA is described in Section II. Section III presents the simulation results of the designed Doherty PA and comparison results with other papers. Finally, a conclusion is given in Section IV.

## II. DESIGN OF THE DOHERTY PA

There are two generic Doherty PA architectures, the current-combined Doherty PA and the voltage-combined Doherty PA, with corresponding load modulation at different input drive voltages. To obtain high output power, the optimum load impedance  $R_o$  is designed to be small. The impedance  $R_o$  is transformed to  $R_o/2$  in the current-combined Doherty PA while  $2R_o$  in the voltage-combined Doherty PA [8]. Compared to the current-combined Doherty PA, the voltage-combined Doherty PA has a lower impedance transformation ratio, which results in a large bandwidth and eases the output matching network design. The voltage combining method is used in this paper.

The differential structure is used in this paper, which enables the differential signal to be amplified and the common-mode signal to be suppressed. The even harmonics, the noise on the supply and the noise on the substrate can be effectively suppressed in the differential circuit. In the standard RF CMOS process, there is no ground via that perfectly connects the transistor's source to ground. The differential structure is capable of providing a better virtual ground and the effect of bonding wires on amplifier performance during chip packaging can be offset by the differential structure. In addition, the differential structure is used to increase the output voltage swing compared to the single-ended structure, thus increasing the amplifier output power.

The architecture of the Doherty PA, shown in Fig. 1, consists of a phase compensate circuit, an input matching network, two amplifiers including the main amplifier and auxiliary amplifier (aux. amplifier), an impedance inverter and a transformer-based voltage combining network. The main amplifier usually works in class AB while the aux. amplifier works in class C. The phase shift generated by the impedance

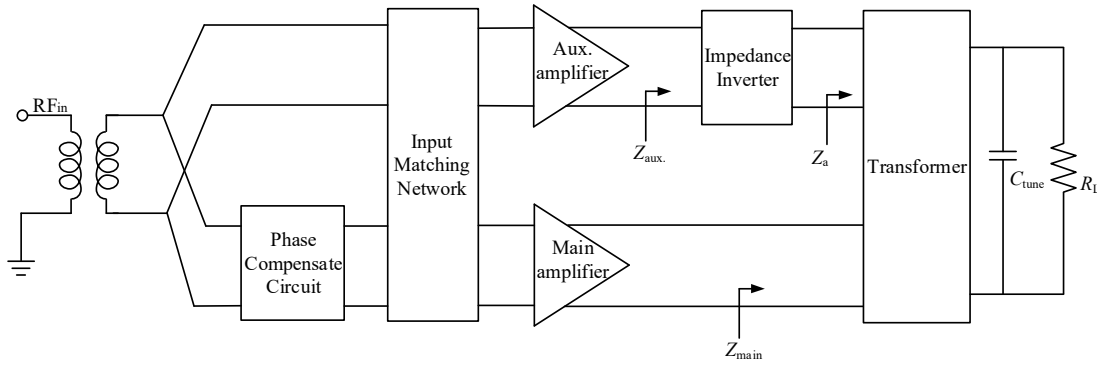


Fig. 1. Block diagram of the Doherty PA.

inverter is compensated by the phase compensate circuit.

As shown in Fig. 1, the input signal is from the left port, then flows through an off-chip balun to convert the single-ended signal to a differential signal. In the upper path, the differential signal flows into the aux. amplifier through the input matching network, and then flows into the transformer through the impedance inverter. Meanwhile, in the latter path, the differential signal flows through the phase compensate circuit, the input matching network and the main amplifier in sequence, then flows into the transformer together with the output signal of the impedance inverter. Finally, the attached capacitor is used to tune the impedance.

The quarter-wave transmission line is utilized at the output of the aux. amplifier to achieve impedance inversion. But at 2.4 GHz, a larger area is required by using the quarter-wave transmission line, a lumped element  $\pi$  network instead of the quarter-wave transmission line is used in this paper. In the phase compensate circuit, lumped element  $\pi$  networks are also used instead of the quarter-wave transmission lines to compensate the phase shift. The transformer is a good choice to achieve voltage combining. In addition to voltage combining, the transformer can also provide a low loss matching circuit.

The diagram of the Doherty PA based on the transformer is shown in Fig. 1. The output impedance of the transformer can be expressed as

$$Z_{out} = 2Z_a + 2Z_{main} \quad (1)$$

In low power region, the voltage is not enough to turn on the aux. amplifier, the current flowing through the aux. amplifier is zero. The load impedance seen by the aux. amplifier is infinite, as in:

$$Z_{aux} = \infty \quad (2)$$

The infinite load impedance can be inverted from the infinite to zero by the impedance inverter, resulting in the impedance at the transformer's input being:

$$Z_{main} = 2R_o \quad (3)$$

The ideal voltage and current variation of the Doherty PA is shown in Fig. 2. When the input power gradually increases from zero to half of the maximum input voltage, the output voltage of the main amplifier gradually increases and reaches the maximum voltage  $V_{max}$ . The aux. amplifier turns on at half of the maximum input voltage, and the current is increased from zero to continuously increase the output power. As the input power continues to increase, the output voltage of the main amplifier can remain maximum to maintain the peak efficiency. The voltage and current of the aux. amplifier can increase continuously. Both amplifiers reach their peak efficiency and the second peak in the overall efficiency can

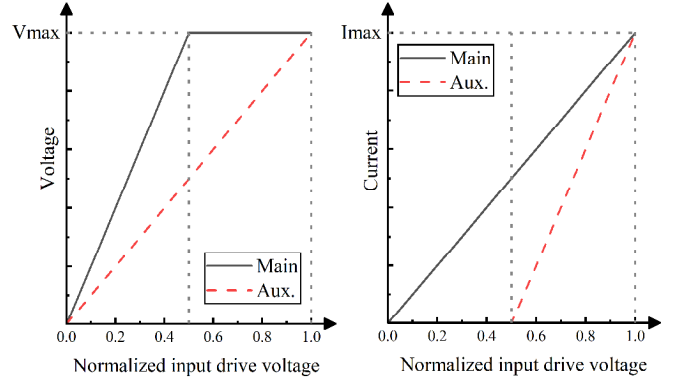


Fig. 2. Ideal voltage and current variation of the Doherty PA.

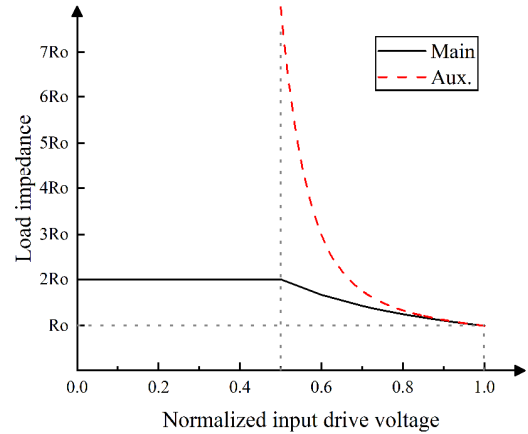


Fig. 3. Ideal load impedance variation of the Doherty PA.

be obtained when the output voltage of the aux. amplifier also reaches its peak value. In this process, the load impedance seen by the aux. amplifier changes from the infinite to  $R_o$ , while the load impedance seen by the main amplifier changes from  $2R_o$  to  $R_o$ . The load impedance variation curve is shown in Fig. 3.

Fig. 4 depicts the schematic of the designed Doherty PA. The differential cascode structure is utilized in the designed Doherty PA. The common source is selected with thin-oxide type transistors to provide high gain. The thin-oxide type transistor has a high cut-off frequency, but it has a reliability issue. The common gate is selected with thick-oxide type transistors. Although the thick-oxide type transistor cannot provide enough gain, it cannot pose a reliability issue. It is advantageous to get high gain without a reliability issue by using the cascode structure. 3.3 V supply voltage is used while the common gate is biased at 1.9 V. And the gate voltages of

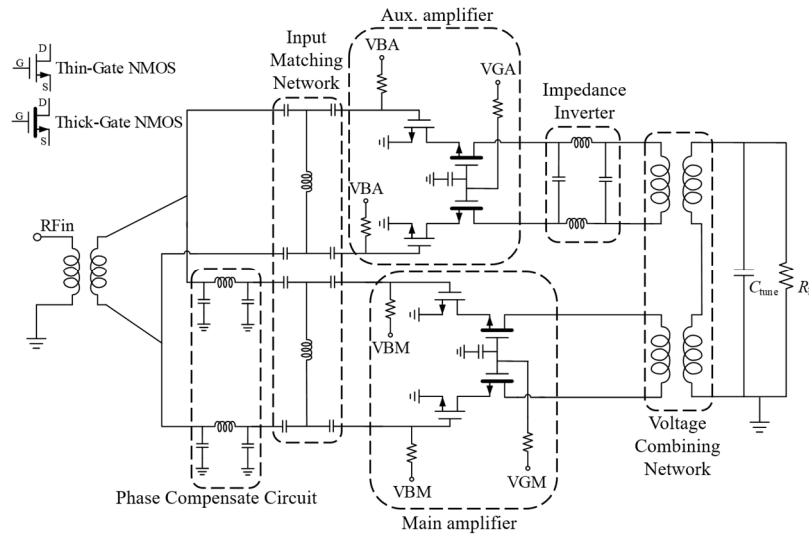


Fig. 4. Schematic of the designed Doherty PA.

the common source transistors in the main and the aux. amplifiers are set to 0.35 V and 0.22 V, respectively.

In the Doherty PA, the main amplifier usually works in class AB, which has lower efficiency and higher linearity. The aux. amplifier usually works in class C, which has higher efficiency and lower linearity. In this paper, linearity and efficiency are considered in a compromise. In order to get lower third order intermodulation distortion (IMD3) at 9 dB and 12 dB PBOs, the asymmetric structure is used in this paper to make the current of the main amplifier much larger than the aux. amplifier. In addition, the existence of parasitic resistance cause the phase compensate circuit to be lossy, so the asymmetric Doherty PA with different sized devices is designed in this paper.

### III. RESULTS AND COMPARISON

The layout of the Doherty PA in 22 nm CMOS technology is shown in Fig. 5. The total size including all the pads is  $1.02 \times 1.07 \text{ mm}^2$ .

The secondaries of the transformer are connected in series for voltage combining. To minimize losses and satisfy dc electro-migration requirements, the primary and the secondary are routed in the top three thicker layers. The maximum line width allowed in the design rules is used in the transformer. Multiple metal layers are connected in parallel at the center tap of the transformer to reduce parasitic resistance. Meanwhile, in order to improve the quality factor of the transformer, the inner radius is increased as much as possible, which can effectively improve the power added efficiency (PAE) of the designed Doherty PA.

The post-simulated gain and efficiency of the Doherty PA at 2.4 GHz are plotted in Fig. 6 together with the current of the main and the aux. amplifiers. As the post-simulation results show, the Doherty PA achieves 30.7 dBm saturated output power ( $P_{\text{sat}}$ ) at 2.4 GHz and has a gain of 20.1 dB. The 1 dB output power compression point (OP1dB) is 28.1 dBm and the peak PAE is 42.5 %. The Doherty PA achieves 25.2 % PAE at 6 dB PBO, achieving significant PBO efficiency enhancement. The PAE of the designed Doherty PA is 16.2 % at 9 dB PBO and 9.6 % at 12 dB PBO. When the output power is above 24 dBm, the aux. amplifier starts to turn on and the power generated from the aux. amplifier is used to compensate the saturated power generated from the main

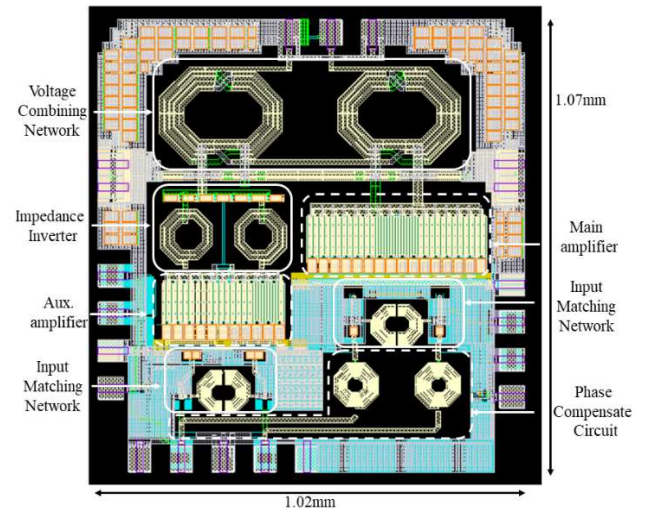


Fig. 5. Layout of the Doherty PA.

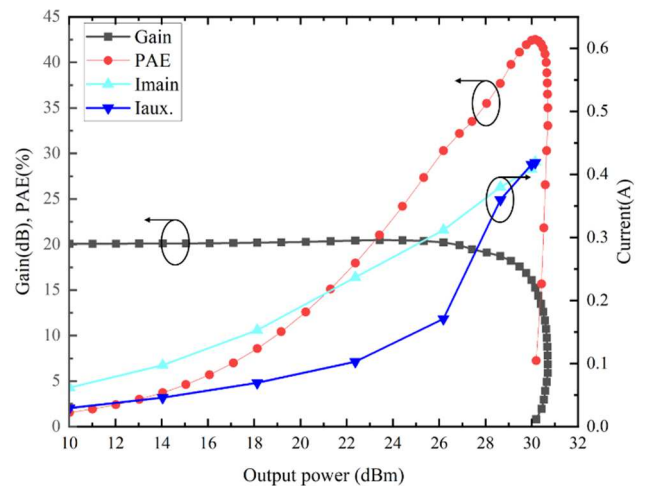


Fig. 6. Gain, efficiency and current versus output power.

amplifier. At the maximum output power point, both amplifiers flow almost the same current and generate the same output power.

A two-tone test is performed at 2.4 GHz with a tone

TABLE I. COMPARISON OF 2.4 GHz CMOS PAs

| Reference   | Architecture                    | Output matching | CMOS process       | Supply (V) | $P_{\text{sat}}$ (dBm) | OP1dB (dBm) | Peak Efficiency (%) | Efficiency @ 6 dB PBO (%) | IMD3 @ 9 dB PBO (dBc) | IMD3 @ 12 dB PBO (dBc) |
|-------------|---------------------------------|-----------------|--------------------|------------|------------------------|-------------|---------------------|---------------------------|-----------------------|------------------------|
| [ 9 ]*      | Quadrature SFCPA                | transformer     | 40 nm              | 1.2/2.4    | 30.3                   | N.A.        | 41.3 (DE)           | 36.1 (DE)                 | N.A.                  | N.A.                   |
| [ 10 ]*     | Multi-mode outphasing           | transformer     | 45 nm              | 2.4        | 31.6                   | N.A.        | 49.2 (DE)           | 25.3 (DE)                 | N.A.                  | N.A.                   |
| [ 11 ]*     | 3-stage differential PA         | LC network      | 65 nm              | 3.3        | 31.5                   | 27.5        | 25 (PAE)            | 14*** (PAE)               | N.A.                  | N.A.                   |
| [ 12 ]*     | 2-stage differential Doherty PA | transformer     | 90 nm              | 2          | 20.5                   | 17.5        | 26.7 (PAE)          | 15*** (PAE)               | − 36.1***             | − 40***                |
| [ 13 ]**    | 1-stage single-ended Doherty PA | LC network      | 0.13 $\mu\text{m}$ | 2.5        | 24.9***                | 22.9        | 38*** (PAE)         | 17.8 (PAE)                | − 31.5***             | − 44.2***              |
| This work** | 1-stage differential Doherty PA | transformer     | 22 nm              | 3.3        | 30.7                   | 28.1        | 42.5 (PAE)          | 25.2 (PAE)                | − 44.2                | − 55.3                 |

\*Measurement results \*\*Simulation results \*\*\*Graphically estimated

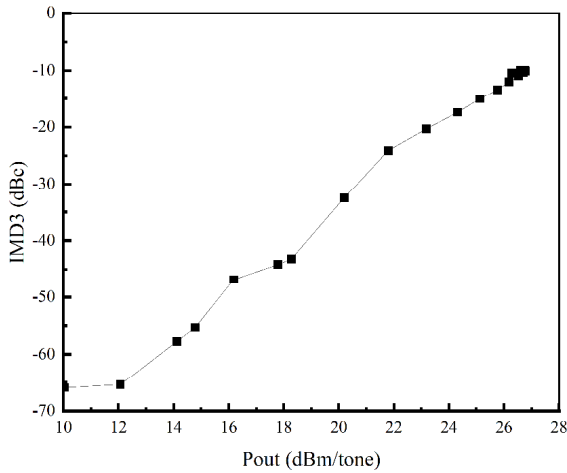


Fig. 7. Two-tone simulated results at 2.4 GHz with 12 MHz tone spacing.

spacing of 12 MHz. The post-simulated IMD3 characteristic versus the output power is shown in Fig. 7. The IMD3 at 9 dB PBO is − 44.2 dBc and at 12 dB PBO is − 55.3 dBc. Ideally, the aux. amplifier is completely closed at 9 dB and 12 dB PBOs. However, due to non-ideal effects, the current of the aux. amplifier cannot be zero. The current of the aux. amplifier should be minimized to ensure that the current of the main amplifier much larger than the aux. amplifier to improve linearity.

Table I summarizes the simulation results of the designed Doherty PA and compares them to the other Doherty PAs. In this work, advanced technology and careful design make the designed Doherty PA have great advantages in high efficiency and high linearity.

#### IV. CONCLUSION

A 2.4 GHz Doherty PA in 22 nm CMOS is designed in this paper. The differential cascode structure is used in the Doherty PA. The lumped element  $\pi$  networks are utilized in the phase compensate circuit and impedance inverter. And the transformer-based voltage combining network is used to achieve load modulation. The Doherty PA designed in this paper with asymmetric structure achieves high efficiency and high linearity. It is suitable for applications requiring high efficiency and linearity.

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